# Journal Pre-proof




Impact of the Sequencing Method on the Detection and Interpretation of Mitochondrial DNA Length Heteroplasmy

Kimberly Sturk-Andreaggi, Walther Parson, Marie Allen, Charla Marshall

PII: S1872-4973(19)30242-X

DOI: https://doi.org/10.1016/j.fsigen.2019.102205

Reference: FSIGEN 102205

To appear in: *Forensic Science International: Genetics*

Received Date: 31 May 2019

Revised Date: 9 November 2019

Accepted Date: 9 November 2019

Please cite this article as: Sturk-Andreaggi K, Parson W, Allen M, Marshall C, Impact of the Sequencing Method on the Detection and Interpretation of Mitochondrial DNA Length Heteroplasmy, *Forensic Science International: Genetics* (2019), doi: https://doi.org/10.1016/j.fsigen.2019.102205

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2019 Published by Elsevier.

Journal Pre-proof

# Impact of the Sequencing Method on the Detection and Interpretation of Mitochondrial DNA Length Heteroplasmy

**Kimberly Sturk-Andreaggi**[1,2,5]✉, **Walther Parson**[3,4], **Marie Allen**[5], **Charla Marshall**[1,2,4]

[1]*Armed Forces Medical Examiner System, Armed Forces DNA Identification Laboratory, Dover, Delaware, United States*
[2]*ARP Sciences, LLC, Rockville, Maryland, USA*
[3]*Institute of Legal Medicine, Medical University of Innsbruck, Innsbruck, Austria*
[4]*Forensic Science Program, The Pennsylvania State University, University Park, Pennsylvania, USA*
[5]*Department of Immunology, Genetics and Pathology, Science for Life Laboratory, Uppsala University, 751 08 Uppsala, Sweden.*

✉corresponding author: Armed Forces Medical Examiner System, Armed Forces DNA Identification Laboratory, 115 Purple Heart Drive, Dover, Delaware, USA; kimberly.s.andreaggi.ctr@mail.mil, +1 302 346-8520

**Highlights**

- Sanger, Illumina and Ion sequencing methods generated mitochondrial DNA data
- Length heteroplasmy in polycytosine tracts of hypervariable regions 1 and 2 was characterized
- Patterns of length heteroplasmy was comparable between the Sanger and Illumina data
- More length variation was observed in the Ion data due to increased indel error
- Enrichment and analysis strategies also impacted length heteroplasmy

## Abstract

Advancements in sequencing technologies allow for rapid and efficient analysis of mitochondrial DNA (mtDNA) in forensic laboratories, which is particularly beneficial for specimens with limited nuclear DNA. Next generation sequencing (NGS) offers higher throughput and sensitivity over traditional Sanger-type sequencing (STS) as well as the ability to quantitatively analyze the data. Changes in sample preparation, sequencing method and analysis required for NGS may alter the mtDNA haplotypes compared to previously generated STS data. Thus, the present study aimed to characterize the impact of different sequencing workflows on the detection and interpretation of length heteroplasmy (LHP), a particularly complicated aspect of mtDNA analysis. Whole mtDNA genome (mitogenome) data were generated for 16 high-quality samples using well-established Illumina and Ion methods, and the NGS data were compared to previously-generated STS mtDNA control region data. Although the mitogenome haplotypes were concordant with the exception of length and low-level variants (<30% variant frequency), LHP in the hypervariable segment (HVS) polycytosine regions (C-tracts) differed across sequencing methods. Consistent with previous studies, LHP in HVS1 was observed in samples with nine or more consecutive cytosines (Cs) and eight Cs in the HVS2 region in the STS data.

Journal Pre-proof

The Illumina data produced a similar pattern of LHP as the STS data, whereas the Ion data were noticeably different. More complex LHP (i.e. more length molecules) was observed in the Ion data, as length variation occurred in multiple homopolymer stretches within the targeted HVS regions. Further, the STS dominant or major molecule (MM) differed from the Ion MM in 11 (37%) of the 30 regions evaluated and six instances (20%) in Illumina data. This is of particular interest, as the MM is used by many forensic laboratories to report the HVS C-tract in the mtDNA haplotype. In general, the STS MMs were longer than the Illumina MMs, while the Ion MMs were the shortest. The higher rate of homopolymer indels in Ion data likely contributed to these differences. Supplemental analysis with alternative approaches demonstrated that the LHP pattern may also be altered by the bioinformatic tool and workflow used for data interpretation. The broader application of NGS in forensic laboratories will undoubtedly result in the use of varying sample preparation and sequencing methods. Based on these findings, minor LHP differences are expected across sequencing workflows, and it will be important that C-tract indels continue to be ignored for forensic queries and comparisons.

**Keywords:** Mitochondrial DNA; next generation sequencing; length heteroplasmy

## Introduction

Mitochondrial DNA (mtDNA) analysis is an integral tool in forensic cases involving samples with little to no detectable nuclear DNA such as hairs and aged skeletal elements. Historically, Sanger-type sequencing (STS) is used to generate mtDNA data for comparison. The qualitative nature of STS data requires visual inspection by an experienced analyst to determine the mtDNA haplotype, which may be complicated by noise and mixed bases (caused by contamination, DNA damage or authentic heteroplasmy). Length heteroplasmy (LHP) is often observed in conjunction with repeated sequence motifs and homopolymer stretches, further complicating the interpretation of STS data. Molecules of varying length (due to different numbers of the repeated base or unit) migrate simultaneously during capillary electrophoresis, resulting in a loss of resolution downstream of the repetitive region. LHP commonly occurs in the polycytosine tracts (C-tracts) located in the hypervariable segments (HVS) of the mtDNA control region (CR) [1-3]. In fact, Irwin *et al.* found that over half (52%) of more than 5,000 population samples displayed CR LHP in STS data, primarily in the HVS2 C-tracts (45%; nps 300-315) [4]. LHP is also observed in the HVS1 C-tract (nps 16180-16183) when T16189C is present, which is found in 19% of global mtDNA haplotypes and at even higher rates (>25%) in East Asian and Native American populations (https://empop.online; [5]) due to the frequency of mtDNA haplogroup B lineages [6-10]. Additional sequencing efforts, including those targeting the entire mitochondrial genome (mitogenome), have identified at least 10 homopolymer or repetitive regions where LHP has also been observed (Table S1) [11,12]. The recently updated query engine SAM2 [13] of the EMPOP database (https://empop.online; [5]) ignores length variation in a total of 13 regions of the mitogenome including 16193, 291, 309, 315 (when 310 is present), 455, 463, 524, 573, 960, 5899, 8276, 8285, and 8289.

Journal Pre-proof

LHP is thought to result from slippage of the polymerase during replication [1,14-16]. In the case of C-tracts, studies have shown that LHP is observed when seven or more consecutive cytosines (Cs) occur, though the exact number that triggers LHP may be dependent on the region's surrounding motif [4,17,18]. This is similar to the mechanism that results in stutter products associated with nuclear DNA testing of short tandem repeat (STR) markers [19]. Similarly, stutter of the dinucleotide AC repeats at nps 515-524 of the mtDNA genome is observed as LHP [4,20,21]. Replication slippage can occur *in vivo*, but it also may arise during *in vitro* replication events including PCR amplification. Therefore the fidelity of the enzyme used for replication, in addition to other factors such as cycle number, may induce or exacerbate LHP in homopolymer or STR regions [14,16,22,23]. However, studies evaluating various PCR conditions have shown that cycle number and the type of polymerase had little impact on LHP [24,25]. Inconsistent displays of LHP were observed across replicates of the same sample in other studies [2,24]. Additionally, variation in LHP has been observed across tissue types from the same individual [2,26-28] as well as within family lineages [29-32]. Each of these previous studies demonstrated variability in length variants when utilizing STS, and therefore interpretation guidelines recommend that insertions and deletions (indels) in the CR C-tracts should be ignored in direct forensic comparisons and database searches [33]. However, the observed variability in LHP in STS data could potentially be attributed to the sequencing chemistry itself. STS uses BigDye Terminator chemistry with Taq FS as the cycle sequencing polymerase, which can lead to more slippage-related errors than the high fidelity polymerases typically used for next generation sequencing (NGS) technologies [23].

Now that NGS methods are widely available, it is worth revisiting the stability of LHP and how it is impacted by sequencing chemistry. The two primary NGS chemistries utilized for mtDNA sequencing are Illumina's sequencing by synthesis (SBS) and Thermo Fisher Scientific's Ion Torrent semiconductor sequencing. The difference in chemistry between these two NGS platforms could have an impact on the display of LHP. Data generated with Illumina's single-base sequencing method tends to have base misincorporation errors at a rate of 0.1 substitutions per 100 bases [34]. LHP would be most impacted by indel errors, which are detected infrequently in Illumina MiSeq data (<0.001 indels per 100 bases) [34]. In contrast, the dominant sources of error in Ion Torrent semiconductor sequence data (similar to 454 pyrosequencing data [35]) are spurious indels, particularly in homopolymeric regions [36]. A study by Loman *et al.* detected 1.5 indel errors for every 100 bases in Ion Torrent Personal Genome Machine (PGM) data. The Ion Torrent semiconductor sequencing chemistry is dependent on the change in pH, which is converted to an electronic signal, in order to determine the number of nucleotides added during a single base sequencing cycle. However, as the number of nucleotides added per cycle increases, the signals of sequences with similar homopolymer lengths (e.g., seven and eight Cs) become difficult to differentiate. In fact, Loman *et al.* noted accuracy rates as low as 60% in homopolymer regions of six bases or more [34]. Based on the higher frequency of deletion errors observed in Ion data [37], homopolymer lengths are likely to be underestimated, which may impact the length variants detected and ultimately the mtDNA haplotype. Although sequencing error studies to date have characterized the earlier Ion platforms (namely the PGM) (e.g. [34,36]), the Ion Torrent semiconductor technology is shared by all Ion platforms. Since both the Illumina and Ion platforms are being marketed for mtDNA sequencing in forensic laboratories, it

Journal Pre-proof

will be important to characterize the effect of the sequencing chemistry on the reporting of LHP in mtDNA profiles. In addition to alternative sequencing chemistries from which to evaluate LHP, NGS offers the ability to quantify heteroplasmic length variants with greater reliability [38,39], potentially increasing mtDNA haplotype resolution.

The present study aimed to characterize the impact of sequencing chemistry on the display of LHP and to determine the validity of utilizing LHP in mtDNA comparisons in forensics. Thus, HVS1 and HVS2 C-tract data from gold standard STS analyses were compared to NGS data generated with Illumina and Ion sequencing. The Illumina data were derived from two long-range (LR) PCR products from high-quality DNA samples [40], while Ion data were generated using a small-amplicon (SA) PCR approach [41] that is suitable for all types of forensic samples including degraded casework specimens. The NGS workflows were selected for this study as they are commonly used in forensics for mitogenome sequencing (e.g. [42-50]). These well-established NGS methods offer a quantitative analysis of mitogenome sequence data, which allowed for an investigation of the effects of the sequencing platform as well as enrichment strategy on LHP.

## 1. Materials and Methods

### *1.1. Samples*

Sixteen anonymized serum samples from the Department of Defense Serum Repository were selected to represent a range of HVS1 and HVS2 C-tract lengths based on whole CR STS data, which were previously generated using the methods described in Scheible *et al.* [51] (Table 1). DNA was extracted with the QIAamp 96 DNA Blood Kit (QIAGEN, Hilden, Germany) on either QIAGEN 9604 or Hamilton STARlet (Reno, NV, USA) automated platforms. Extracts were quantified with Plexor HY System (Promega, Madison, WI, USA) on an Applied Biosystems 7500 Real-Time PCR System (Thermo Fisher Scientific, Waltham, MA, USA). The use of these samples was reviewed by the Institutional Review Board Office under the U.S. Army Medical Research and Material Command's Office of Research Protections, and it was determined not to involve human subjects.

Table 1. The major molecule reported for the HVS1 and HVS2 C-tracts based on previously generated Sanger-type sequencing (STS) data. Observation of minor to moderate length heteroplasmy is denoted with a "*" following the motif and the "**" indicates that severe length heteroplasmy was observed.

| Sample | HVS1 | HVS2 |
|---|---|---|
| 1 | 16182C 16183C 16189C 16193.1C** | 309.1C 309.2C 309.3C 315.1C* |
| 2 | 16189C 16192T | 309.1C 315.1C* |
| 3 | rCRS | 310C 315-** |
| 4 | rCRS | 309.1C 309.2C 315.1C* |
| 5 | 16185T 16189C 16193- | 309.1C 315.1C* |

Journal Pre-proof

| | | |
|---|---|---|
| 6 | 16189C** | 309.1C 309.2C 315.1C* |
| 7 | rCRS | 310C** |
| 8 | 16189C 16191.1C 16192T* | 309.1C 315.1C* |
| 9 | 16187T 16189C | 309.1C 315.1C* |
| 10 | 16182C 16183C 16189C 16193.1C 16193.2C** | 310C 313- 314- 315-** |
| 11 | 16183C 16189C 16193.1C** | 315.1C 315.2C |
| 12 | 16183C 16189C** | 315.1C |
| 13 | 16182C 16183C 16188T 16189C 16193.1C 16193.2C | 309- 315.1C |
| 14 | 16182C 16183C 16189C 16193.1C** | 308- 309- 315.1C |
| 15 | 16181G 16182C 16183C 16189C** | 309- 315.1C |
| 16 | 16189C 16193.1C 16193.2C** | 309.1C 309.2C 315.1C* |

### *1.2. STS data generation and analysis*

STS data targeting the HVS1 and HVS2 C-tract regions were generated by first amplifying the entire CR (Table S2). PCR amplification using AmpliTaq Gold DNA Polymerase (Thermo Fisher Scientific) was performed as described previously in [52]. PCR products were quantified with the DNA 7500 Kit (Agilent, Santa Clara, CA, USA) on a 2100 Bioanalyzer (Agilent) and then purified with a 0.6X AMPure XP (Beckman Coulter, Indianapolis, IN, USA) prior to STS. The sequencing reactions contained 2 µL purified PCR product, 4 µL BigDye Terminator v1.1 Ready Reaction Mix (Thermo Fisher Scientific), 2 µL BigDye Terminator v1.1 Sequencing Buffer (Thermo Fisher Scientific), 1 µL sequencing primer (10 µM) and 11 µL water. Four sequencing reactions were performed for each sample in order to cover the targeted C-tracts with forward and reverse primers: F15971 and R16410 for HVS1, and F155 and R484 for HVS2 [53]. Cycling was performed on a GeneAmp 9700 PCR System (Thermo Fisher Scientific) according to the manufacturer's protocol. Sequencing products were purified with the Performa DTR 96-Well Ultra Plate (Edge BioSystems, Maumee, OH, USA), dried down, and re-suspended with 10 µL Hi-Di Formamide (Thermo Fisher Scientific) before capillary electrophoresis on an Applied Biosystems 3500xl Genetic Analyzer (Thermo Fisher Scientific).

STS data were analyzed in Sequencher version 5.0 software (GeneCodes, Ann Arbor, MI, USA). Sequences were trimmed manually and then aligned to the revised Cambridge Reference Sequence (rCRS; [54,55]). The HVS C-tracts were assessed in accordance with the DNA Commission of the International Society for Forensic Genetics (ISFG) recommendations for LHP regions in reference data [33], in which the dominant length variant, or major molecule (MM), is to be reported. The MM in the STS data was determined by two scientists, and the haplotype associated with the highest single, non-repetitive nucleotide downstream of the LHP was reported. The single guanines (Gs) following each of the HVS1 and HVS2 C-tracts (i.e. G16196 and G316, respectively) were used to make this determination. The sequence of the MM was then converted into its corresponding motif and represented in the sample profile. For example, if the HVS2 MM sequence is 300-AAACCCCCCCCTCCCCCCG-316 (3A-8C-T-6C compared to 3A-7C-T-5C present in the rCRS), then 309.1C 315.1C was reported in the sample

Journal Pre-proof

haplotype. In addition to the MM determination, HVS1 and HVS2 LHP was further characterized by recording the rCRS-coded haplotype based on the sequence, length (i.e. the number of Cs), and relative proportion of all length molecules detected. Based on complexity (severity) of LHP observed in the original STS data [51], samples were grouped into three categories: no LHP (1 length molecule, 100% MM proportion), minor to moderate LHP (≥50% MM proportion, 2-4 length molecules), and severe LHP (<50% MM proportion, 4+ length molecules) (Table 1).

### *1.3. Illumina data generation*

The mitogenomes for the 16 samples were sequenced on an Illumina MiSeq FGx Forensic Genomics System (San Diego, CA, USA) according to the procedures described in Ring *et al.* [56]. Enrichment PCR was performed with two overlapping, long-range (LR; ~8500bp) primer sets and Advantage GC Genomic LA DNA Polymerase Mix (Takara Bio USA Inc., Mountain View, CA, USA) to target the mitogenome. LR amplicons were pooled, purified and quantified prior to library preparation. Each purified amplicon pool (50 ng) was fragmented and prepared for Illumina sequencing using a half-volume KAPA HyperPlus (Roche, Indianapolis, ID, USA) procedure [56]. HyperPlus libraries were pooled by equal volume and quantified with the KAPA Library Quantification Kit (ROX Low) for Illumina Platforms (Roche) on a 7500 System. The pool was prepared according to the manufacturer's recommendations, and 12 pM was loaded on a MiSeq FGx system for 150-cycle single-end sequencing using a v3 MiSeq Reagent Kit (Illumina). Additional processing information is presented in Table S2.

### *1.4. Ion data generation*

Small-amplicon (SA) PCR with the Applied Biosystems Precision ID mtDNA Whole Genome Panel (Thermo Fisher Scientific) was used to enrich samples for the mitogenome for Ion sequencing. The Precision ID libraries were prepared with the Ion AmpliSeq Kit for Chef DL8 (Thermo Fisher Scientific) on the Ion Chef (Thermo Fisher Scientific) instrument. Two pools (eight samples each) were prepared by loading 15 µL of each extract onto the instrument and utilizing the 2-in-1 method and 22 cycles recommended by the manufacturer. The resulting library pools were quantified with the Ion Library TaqMan Quantitation Kit (Thermo Fisher Scientific) on a 7500 System. Based on the quantitation results, each pool was diluted to 30 pM. Templating of an Ion 520 chip was performed using the Ion Chef, and then each pool/chip was sequenced on the Ion S5 according to the manufacturer's recommendations. An overview of the processing workflow for SA Ion NGS analysis is presented in Table S2.

### *1.5. Supplemental data generation*

To investigate whether the observed differences in LHP were due to the sequencing chemistry or the enrichment process, supplemental data were generated for comparison. First, purified CR and LR amplicons were sequenced with the alternate sequencing methods (i.e. Illumina and STS, respectively; Table S2).Secondly, CR products using the same amplification primers (F15971 and R599) were generated using the Q5 Hot-Start High-Fidelity DNA Polymerase (New England Biolabs Inc., Ipswich, MA, USA), which has a fidelity >200 times greater than Taq [23]. The CR-Q5 products were processed with STS, as well as sequenced on the Illumina MiSeq, using

Journal Pre-proof

the same methods as the other CR (i.e. AmpliTaq Gold) products (Table S2). Lastly, the SA Ion library pools were prepared for Illumina sequencing using the KAPA HyperPrep Kit (Kapa Biosystems) and sequenced on the MiSeq FGx System (Table S2) to directly compare Ion and Illumina data.

### *1.6. NGS data analysis*

The FASTQ files generated by the Illumina MiSeq runs were imported into the CLC Genomics Workbench v7.5.1 software (QIAGEN) and analyzed with a custom workflow (Supplemental File 1). Sequences were trimmed and then mapped to a circular version of the rCRS using optimized parameters. BAM files (aligned to a modified rCRS+80 bp reference genome [44]) produced by the S5 data by the Ion Torrent Variant Caller plug-in were imported into CLC software, and mitogenome haplotypes were generated using a workflow specific for SA Ion data (Supplemental File 2). Variants were called in both sets of data with the Low Frequency Variant Detection tool at positions with a read depth of at least 100X. Heteroplasmy was reported when the minor nucleotide was observed in at least five reads *and* was present at a frequency of 5% or higher. Additional variant calling parameters, specific to the sequencing method, were employed in the respective workflows to ensure accurate haplotype generation (Supplemental Files 1-2). The haplotypes were then modified based on forensic conventions using the CLC AQME plug-in [57], and manual edits to the profile were made as necessary. Haplogrouping was performed with AQME's Mitochondrial Haplogrouper tool based on PhyloTree Build 17 [58,59], and profiles were updated to align indels in accordance with the associated phylogeny [33,60]. Resulting mitogenome profiles for each sample were compared across the two NGS strategies, as well as CR STS data, to ensure concordance.

Supplemental haplotype analyses of the SA Ion BAM files were performed using a custom IGV mtDNA tool (variant caller v1.01b) that was developed specifically for Ion sequencing of the Precision ID mtDNA Panels [42,61]. Median read depth was determined for each amplicon, and a 20X minimum read depth threshold was used for variant detection. Default variant calling parameters were utilized, including 20 reads to call a variant with a minimum variant frequency of 10% for substitutions, 20% for insertions, and 30% for deletions. The tool automatically flags and removes variants associated with nuclear mtDNA segments (NUMTs). These analyses allowed a direct comparison of the user-developed CLC workflow and an optimized SA Ion workflow that was specifically designed for the data type.

### *1.7. AQME LHP analysis*

The custom read count analysis tool of AQME [57] groups mapped reads that span the entire length of a user-specified region (e.g. HVS1 C-tract) by sequence and then reports the number of reads per group. The AQME LHP analysis is not impacted by alignment of reads within a mapping, but mapping parameters and reference bias may impact LHP by affecting the reads included in the assembly [37,62]. The analysis workflows applied the AQME read count tool in this study for two homopolymeric regions: HVS1 (nps 16180-16194) and HVS2 (nps 300-316) to include the polyadenine stretches (A-tracts) prior to each of the C-tracts. Reads that extended across the entire specified region were grouped by sequence, and those sequences with five or

Journal Pre-proof

more (≥5X) reads were reported. The read count tables were then exported as Microsoft Excel files (Redmond, WA, USA), and regions with fewer than 100 reads were identified and excluded. The length of the sequence and the relative proportion within the region were reported for each length molecule. The relative proportion was calculated for each sequence by dividing the number of reads reported for a sequence by the total number of reads identified for all sequences within a region. For example, if 249 reads were classified as a particular sequence (e.g. AAACCCCCCCCCCTCCCCCCG), and a total of 683 reads were identified for the region (e.g. HVS2) using the AQME read count tool, the proportion for this sequence would be recorded as 33.7% (249/683; as in the Table S3 HVS2 example). To further eliminate any sequencing error and negligible length variants, only sequences with a ≥5% proportion of the total reads were used for further analysis. The relative proportion was recalculated considering only those sequences with proportions exceeding 5%. The bracketed motif, rCRS-coded haplotype, and longest C-tract were determined for each analyzed sequence (see Table S3 for an example output). The tables were imported in Microsoft Access for determination of the MM and further analysis of the HVS length variants.

### *1.8. STRait Razor LHP verification*

STRait Razor was developed for the analysis of sequenced STRs [63-66] but can be configured for LHP analysis [49]. Using a configuration file created to filter reads, which contained flanking sequences adjacent to the HVS1 and HVS2 C-tracts, STRait Razor functions similar to the AQME read count tool. Yet, STRait Razor can be utilized without the need for a mapped alignment since FASTQ files (opposed to mappings) are used as input. The STRait Razor analysis of the FASTQ files generated by each NGS platform ensured that bias was not introduced by the mapping parameters or reference genome (i.e. a circularized rCRS for the Illumina data and the rCRS+80 for the SA Ion data). The configuration file used in this study was modified slightly from [49] to include A16182 and A16183 (Table S4), which are commonly observed as A-C transversions that extend the HVS1 C-tract. Filtered sequences were grouped by length, which would designate an allele in STR analyses, and the relative proportions of each group within the HVS region were calculated. The bracketed motif, rCRS-coded haplotype, and longest C-tract were inferred from each grouping based on the known HVS motif (see Table S5 for an example output). For example, the expected length of the HVS1 region in the rCRS is 12 (AACCCCCTCCCC), with five Cs in the longest consecutive stretch. A sample with a length of 12 but with two HVS1 substitutions at A16183C and T16189C, known from the STS haplotype, would have 11 consecutive Cs (ACCCCCCCCCCC). Length variation was assumed to occur in the longest C-tract (e.g., nps 303-309 in HVS2), and the STRait Razor sequence output was then used for confirmation. As in the AQME read count analysis, regions with <100 identified reads were excluded, and only groups with a ≥5% proportion were used for analysis. The STRait Razor results were also imported into Microsoft Access for MM determination and LHP analysis.

## 2. Results and Discussion

### *2.1. Profile generation and concordance*

Journal Pre-proof

Full mitogenome profiles were generated with read depth averaging 1,902X and 3,787X using the LR Illumina and SA Ion methods, respectively (Table S6). Similarly, the amplicon median read depth averaged 3,135X according to the IGV analysis of the SA Ion data. One sample (Sample 11) performed poorly overall compared to the other samples, with the lowest read depth in NGS data and with low-quality STS data. This sample had one of the lowest DNA concentrations and inefficient amplification across conditions; therefore it was excluded from all further analyses. Although the average read depth across the mitogenome in the SA Ion data was double that of the LR Illumina data for the 15 analyzed samples, the average number of reads used for AQME read count analysis (length molecules >5%) was comparable between the LR Illumina (1,392 reads) and the SA Ion data (1,319 reads). This is due to the different amplification schemes that resulted in larger inter-amplicon read depth imbalances generated with the SA PCR approach of the Ion workflow, which is evident by the lower median read depth observed for the amplicons targeting the HVS C-tracts (1635X for HVS1 and 946X for HVS2). The LR Illumina data showed a greater average read depth in these targeted regions (Table S6).

Outside regions of LHP, the haplotypes were concordant across all conditions with the exception of low-level variants (<30% variant frequency; Table S7) that were expectedly higher in the SA PCR enrichment approach used to generate the Ion data. These low-level variants can be attributed to incomplete removal of SA PCR primers, Ion-associated indel errors [34], and/or sequences from nuclear mtDNA segments (NUMTs). NUMTs are more likely to be co-amplified using SA PCR compared to LR PCR due to the large number of primers and small amplicons that may target homologous pseudogenes in the nuclear genome [44,61]. The IGV mtDNA tool analyses eased interpretation of the SA Ion data by automatically trimming the SA PCR primers and filtering NUMT-associated variants. Excluding length variants, the IGV mtDNA tool produced mitogenome haplotypes identical to those produced with the CLC analysis of the LR Illumina data (>10% variant frequency).

It should be noted that sequence data, regardless of platform, are strongly affected by the sample preparation strategy (i.e. enrichment method). However, concordant mitogenome haplotypes were obtained in this study using the two NGS workflows. The LHP analyses described below examined the impact of the sequencing platform, as well as the associated enrichment strategy, on the display of mtDNA LHP.

### *2.2. Length heteroplasmy complexity*

The LHP complexity observed in both the original STS and the newly generated CR STS data increased in severity as the number of consecutive Cs in the MM increased (Tables 1-2). This observation is consistent with findings in previous studies [4,18,32,67]. In HVS1, samples with eight Cs or fewer had one detectable molecule, nine Cs displayed minor-moderate complexity, and stretches of 10 or more Cs showed severe LHP. The HVS2 region was slightly different, with no LHP in regions with six or seven Cs, minor-moderate LHP with eight to nine consecutive Cs, and severe LHP was detected in C-tracts with 10 or more Cs. Though the LHP complexity was similar in both CR STS datasets (original [51] and this study), two of the 30 MMs analyzed differed by one C (Figure S1). Both were observed in HVS1 C-tracts with ≥12

Journal Pre-proof

Cs, and the likely cause for the discrepancy is stochastic variation of two similarly proportioned length molecules. Therefore, replicates (CR amplification and STS) of the same samples were consistent in MM with <12 consecutive Cs. The LR Illumina data produced a similar pattern of LHP to that of the STS data, whereas the SA Ion data were markedly different with LHP detected in nearly all samples. All SA Ion samples with HVS1 C-tracts of seven consecutive Cs or more exhibited LHP, and two of four samples with the shortest HVS1 C-tracts (five or six consecutive Cs) exhibited LHP as well. Notably, more than one length molecule was observed in all HVS2 C-tracts of the SA Ion data.

Table 2. The number of samples that display length heteroplasmy (i.e. more than one length molecule), grouped by major molecule (MM) polycytosine (C-) length as determined by this study's control region (CR) Sanger-type sequencing (STS) data for the hypervariable segments (HVS) 1 and 2.

| **STS MM C-length** | **HVS1** | | | | **HVS2** | | | |
|---|---|---|---|---|---|---|---|---|
| | **Sample Count** | **CR STS** | **LR Illumina** | **SA Ion** | **Sample Count** | **CR STS** | **LR Illumina** | **SA Ion** |
| 5 | 3 | 0 | 0 | 2 | 0 | - | - | - |
| 6 | 1 | 0 | 0 | 0 | 3 | 0 | 0 | 3 |
| 7 | 2 | 0 | 0 | 2 | 1 | 0 | 0 | 1 |
| 8 | 1 | 0 | 0 | 1 | 4 | 4 | 3 | 4 |
| 9 | 1 | 1 | 1 | 1 | 3 | 3 | 3 | 3 |
| 10 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 2 |
| 11 | 1 | 1 | 1 | 1 | 0 | - | - | - |
| 12 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 |
| 13 | 2 | 2 | 2 | 2 | 1 | 1 | 1 | 1 |
| 14 | 1 | 1 | 1 | 1 | 0 | - | - | - |

The complexity of length variation, represented by the relative proportions and numbers of molecules observed, was used to further evaluate the impact of the sequencing platform on LHP detection. The CR STS and LR Illumina data generated similar average MM proportions, whereas SA Ion data showed MM proportions as much as two fifths lower (Figure 1). Furthermore, in regions of low LHP complexity, the average number of length molecules in the SA Ion data was nearly double that of the other two methods. The higher rate of indel error in Ion data [34,36] caused length variation in multiple homopolymeric stretches of the targeted HVS C-tract regions (e.g. ≥4 consecutive nucleotides such as the A-tract prior to the HVS1 C-tract or 6 consecutive Cs after T310 in the HVS2 C-tract; Table S8). Length variation in the CR STS and LR Illumina data occurred almost exclusively in C-tracts of ≥7 consecutive Cs (Table S8).

Journal Pre-proof

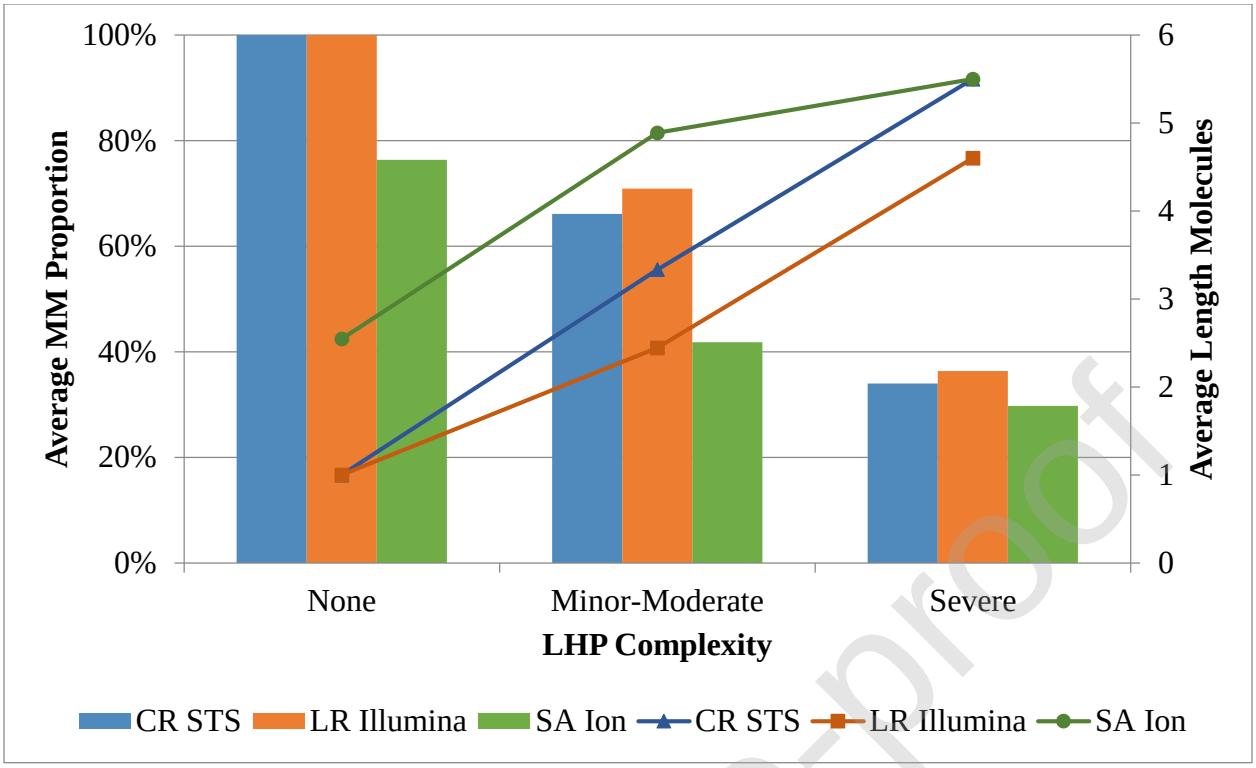



Figure 1. The average proportion of the major molecule (MM) and the number of length molecules based on the complexity of the length heteroplasmy (LHP) observed in the control region Sanger-type sequencing (CR STS), long-range (LR) Illumina, and small-amplicon (SA) Ion data.

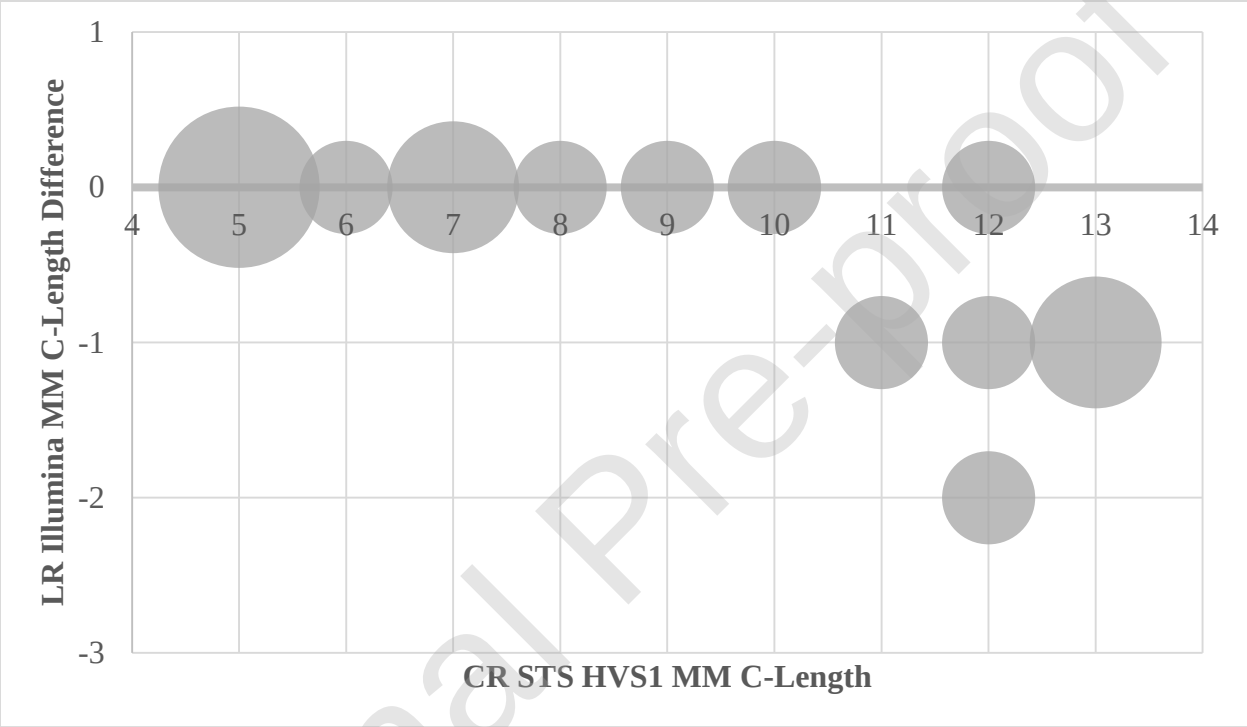
### *2.3. Major molecule comparison*

The complexity of homopolymer regions described above may be of little practical relevance in most forensic laboratories, because the MM is usually reported [33]. For this reason, the determined MMs in the HVS1 and HVS2 C-tracts (30 total regions) were compared across the sequencing methods (i.e. CR STS, LR Illumina, SA Ion). The CR STS MM differed from the LR Illumina MM in six (20%) of the 30 regions evaluated. The difference primarily occurred in the HVS1 C-tract (83%) and always when the CR STS MM had >10 consecutive Cs (Figure 2). Differences between the CR STS and SA Ion MM were observed more frequently (11 out of 30, 37%) and also in stretches of eight Cs (Figure 3). LR Illumina and SA Ion MMs differed in five instances, all of which were located in HVS2 and ≤10 consecutive Cs (Figure 4). When MMs were inconsistent across sequencing platforms, the difference never exceeded two consecutive Cs. Overall, MM C-tract lengths were longer in the CR STS data, followed by LR Illumina and then SA Ion with the shortest MM lengths. Although some of the variability in MM length could potentially be attributed to amplicon size and PCR efficiency, these findings are consistent with expected homopolymer lengths relative to the three sequencing chemistries. Slippage is more likely to occur with the use of the *Taq* DNA polymerase in the CR STS method (AmpliTaq Gold for PCR enrichment and Taq FS for the cycle sequencing reaction), thus resulting in longer MM

Journal Pre-proof

lengths [23]. Shorter MM lengths are expected in the SA Ion data due to homopolymer reduction from the Ion Torrent semiconductor sequencing technology [37]. The LR Illumina method may produce the most accurate MM lengths due to the higher fidelity polymerase (compared to *Taq*) used for the PCR enrichment and the low indel error rate of the MiSeq platform [34]. These results highlight issues that may arise when comparing haplotypes generated with different PCR enrichment and sequencing methods, if homopolymer regions (or any regions of observed LHP) are included in the analyses.

a.




Journal Pre-proof

Journal Pre-proof

b.

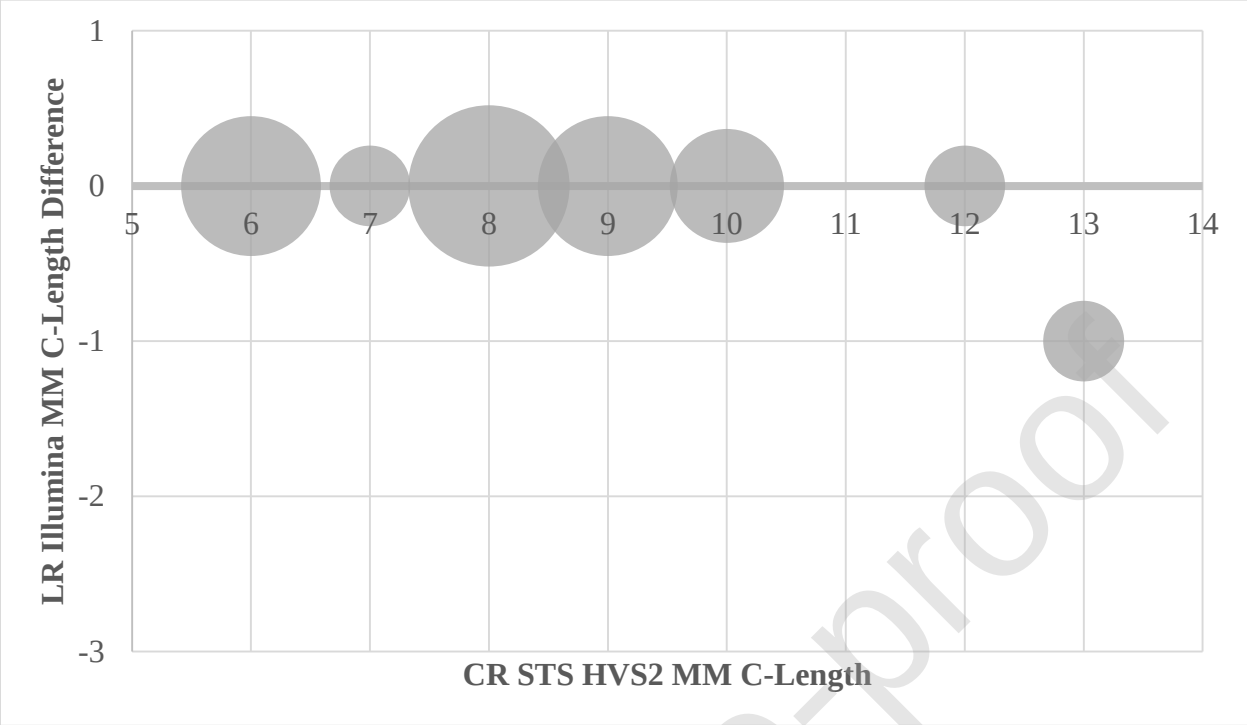
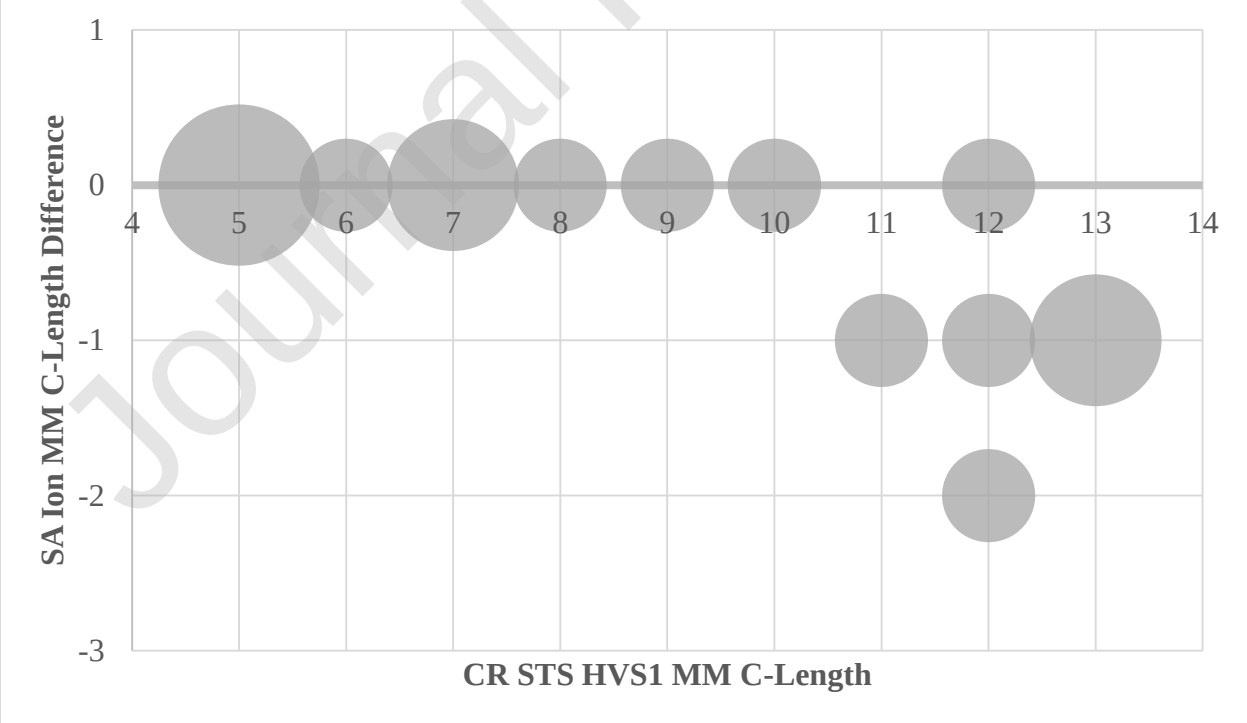



Figure 2. Bubble graphs depicting the difference in major molecule (MM) polycytosine (C-) length between control region (CR) Sanger-type sequencing (STS) and long-range (LR) Illumina data in the a) hypervariable segment (HVS) 1 and b) HVS2 C-tracts.

a.




Journal Pre-proof

Journal Pre-proof

b.

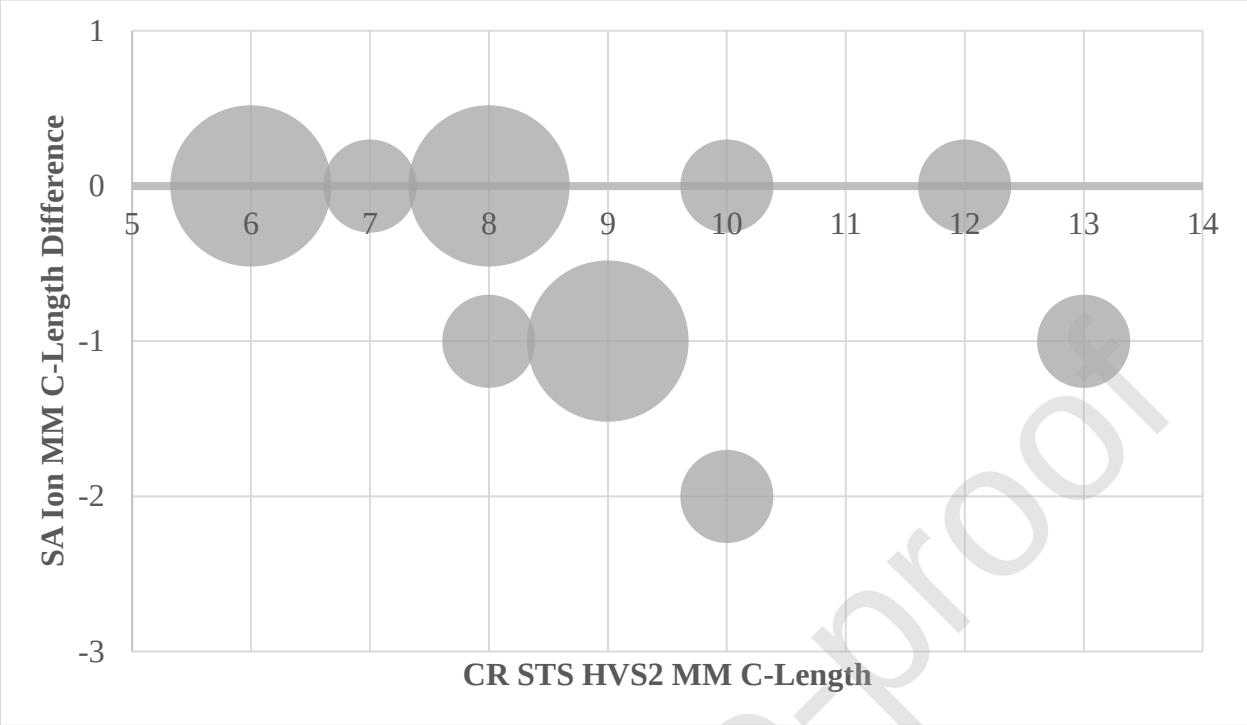
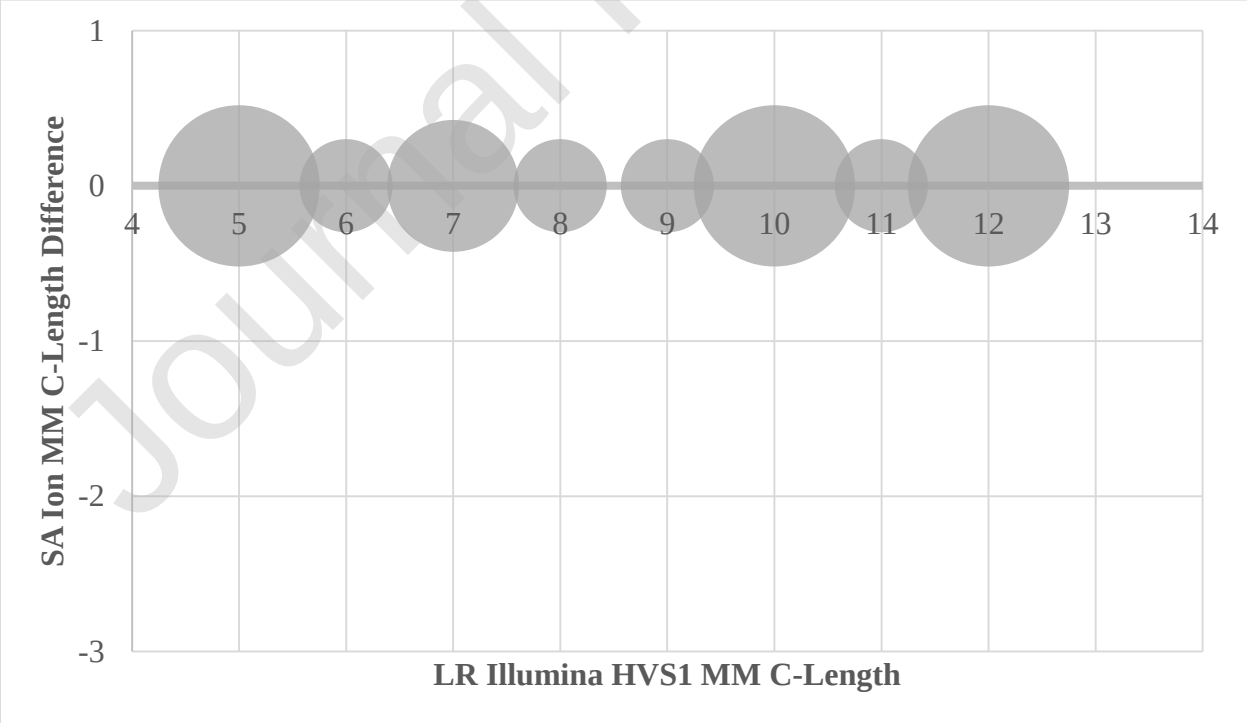



Figure 3. Bubble graphs depicting the difference in major molecule (MM) polycytosine (C-) length between control region (CR) Sanger-type sequencing (STS) and small-amplicon (SA) Ion data in the a) hypervariable segment (HVS) 1 and b) HVS2 C-tracts.

a.




Journal Pre-proof

Journal Pre-proof

b.

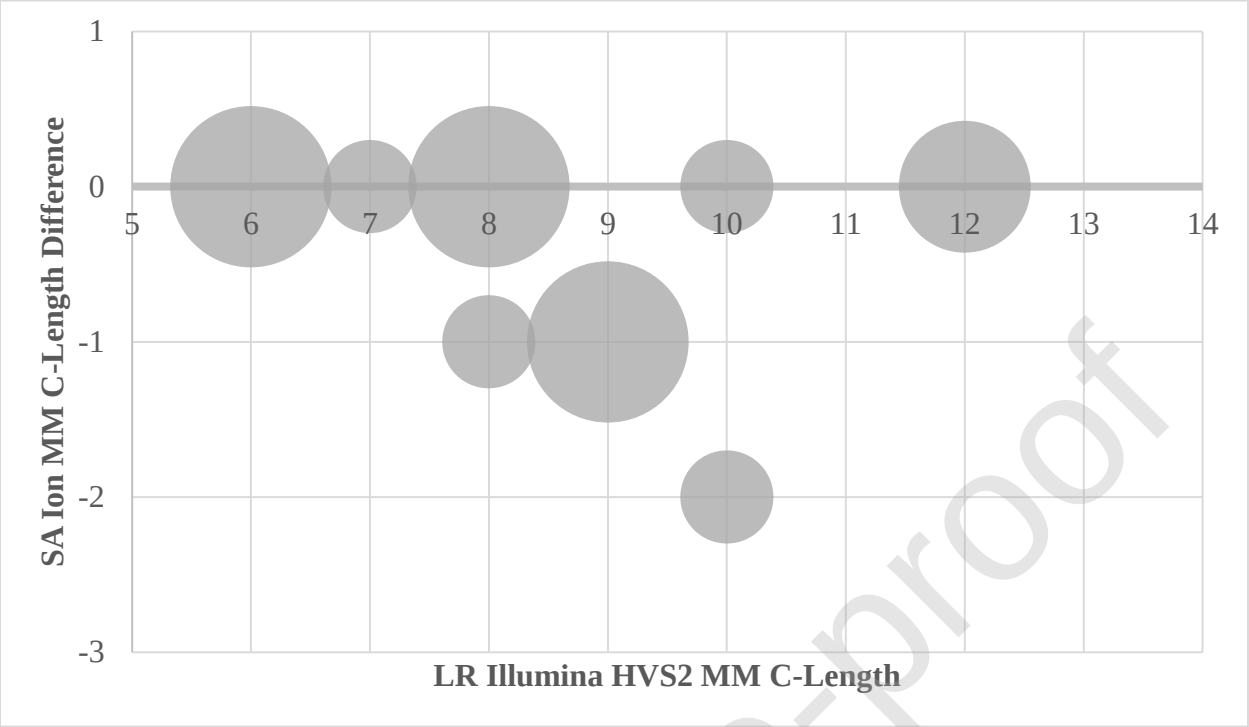



Figure 4. Bubble graphs depicting the difference in major molecule (MM) polycytosine (C-) length between long-range (LR) Illumina and small-amplicon (SA) Ion data in the a) hypervariable segment (HVS) 1 and b) HVS2 C-tracts.

### *2.4. Supplemental data comparison*

The supplemental data generated from the same enriched products were utilized to directly compare sequencing methods. STS data continued to produce longer MM C-tracts than data generated with Illumina sequencing, but only in homopolymer stretches of >10 Cs (Figure S2). This supports the findings presented in Figure 1 and suggests that the sequencing method impacts LHP since the same CR and LR enrichment products were used. This is reinforced by the CR Taq and Q5 results from both STS and Illumina sequencing (Figure S3). Only minimal differences were observed between the conditions and at stretches of 12 or more consecutive Cs, with the exception of one HVS2 region in sample 5. This sample had nearly equal proportions of two molecules of different length (both at ~40%) in all STS and Illumina data (Figure 5); therefore any difference in MM of sample 5 may be due to stochastic variation in DNA template opposed to enrichment and sequencing conditions. For the remaining differences between the CR-Q5 and the other data that were observed at 12 or more consecutive Cs, these may be explained by the Q5 polymerase's higher fidelity that was expected to produce a more accurate replication of the DNA target [14,16,22]. The SA Ion data yielded different patterns of LHP when compared to the STS and Illumina data (including the data from the alternate sequencing method using the same enrichment product). Although the SA Ion MM was consistent with that of the other conditions except for the CR-Q5 data, variation was observed in both HVS2 C-tracts

Journal Pre-proof

(nps 303-309 and 311-315); whereas only the first C-tract in HVS2 varied in length in the non-Ion data. The larger variation observed was likely the result of homopolymer reduction in Ion data [34], which impacts the composition of length molecules and potentially the representation of the LHP (i.e. the motif of the MM) in the profile.

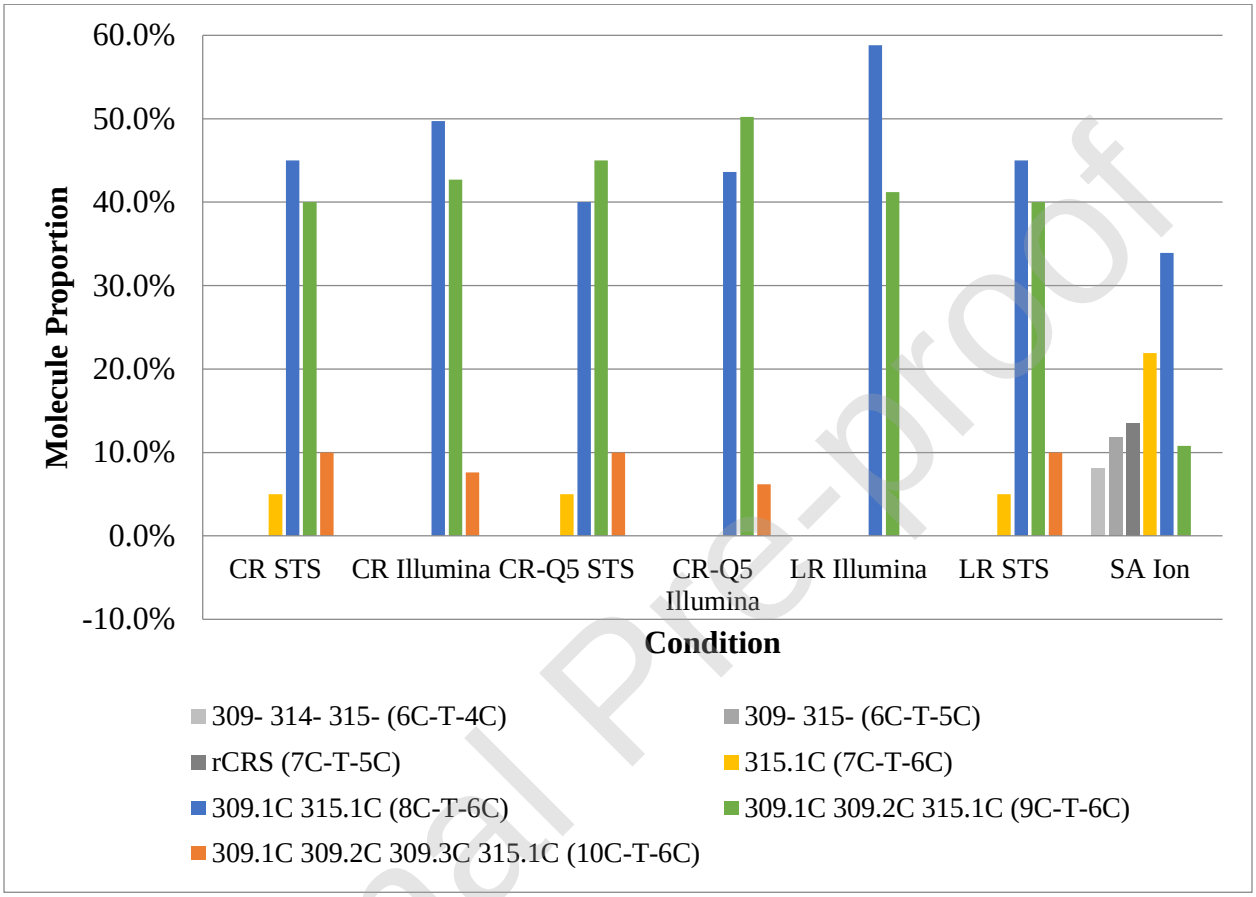



Figure 5. The proportion of length molecules detected in the second hypervariable segment (HVS 2) polycytosine tract of Sample 5.

The direct comparison of Ion to Illumina sequencing from the SA-enriched libraries was limited due to low input into the library preparation kit (<1 ng), which resulted in lower read depth across the mitogenome and notably in the HVS2 region (Table S6 and Figure S4). As a result, nine of the 15 analyzed samples had <100 reads for the LHP analysis of HVS2, but only one HVS1 region with an insufficient read count was observed (Sample 14, 74 reads). When considering the 20 regions that met the 100 read threshold in both datasets, four MM differences were evident between sequencing platforms (Figure 6). The SA Illumina MM was longer than the SA Ion data in three instances, whereas the MM length observed at 12 consecutive Cs in the SA Ion data was shorter in the SA Illumina data. The MM proportion and number of length molecules of the SA Illumina data were consistent with the other Illumina and STS conditions

Journal Pre-proof

and different from the SA Ion in LHP regions of lower complexity (Table S9). These results, from the exact same library product sequenced on both Illumina and Ion platforms, indicate that the sequencing chemistry itself causes variation in LHP.

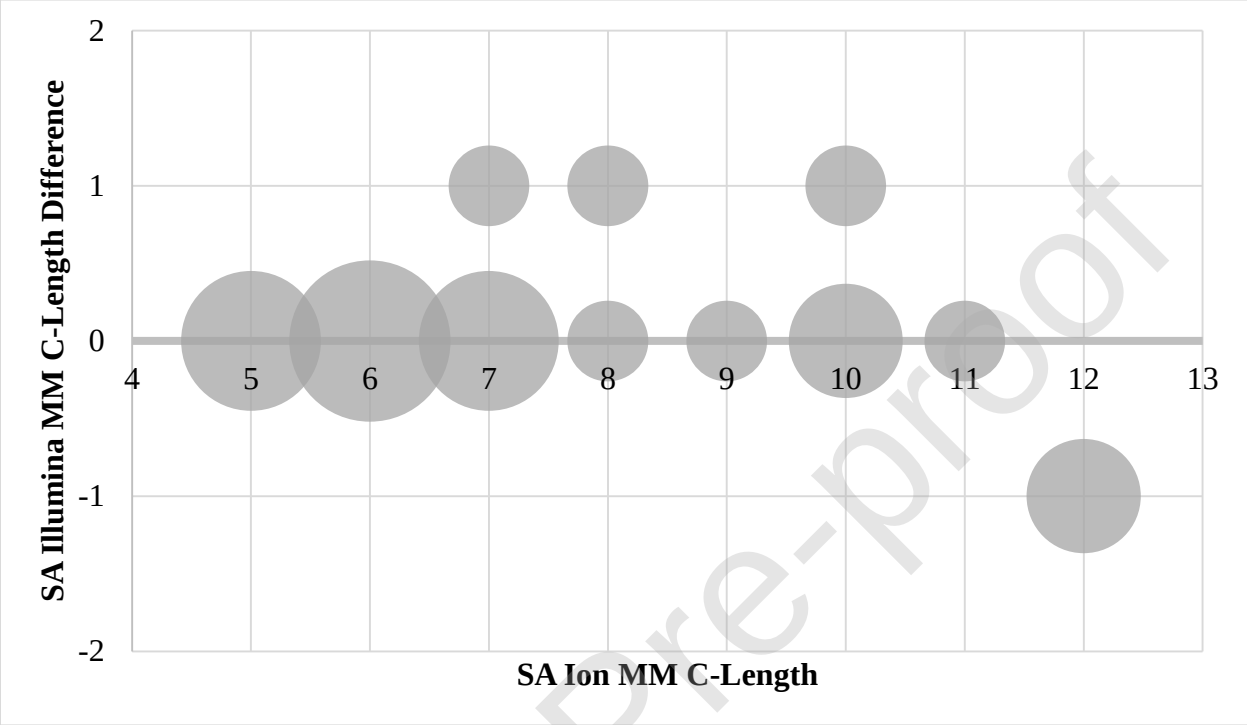



Figure 6. Bubble graphs depicting the difference in major molecule (MM) polycytosine (C-) length between small-amplicon (SA) Ion and SA Illumina data (combines hypervariable segments (HVS) 1 and HVS2 C-tracts).

### *2.5. STRait Razor LHP verification*

Another potential source of LHP variation may be the workflow used to analyze NGS data. When comparing the AQME and STRait Razor results, no differences in MM were observed in the LR, CR, or CR-Q5 Illumina data. However, 10 MM differences were detected in the SA Ion data (Table S10). The four HVS1 MM differences noted between the AQME and STRait Razor LHP analyses were the result of sequences of the same length but different motifs being combined into a single group in the STRait Razor output (Figure S5). When the STRait Razor sequences were inspected, the modified HVS1 MMs were consistent with the AQME results in all HVS1 regions. However, none of the six HVS2 differences between AQME and STRait Razor in the SA Ion data could be clarified by the sequences. One (Sample 3) of the six HVS2 MM differences exhibited more consecutive Cs in the AQME read count analysis of the SA Ion BAM (12 Cs) than the STRait Razor-determined MM (11 Cs in both length and sequence outputs). In this instance, both length molecules (11 and 12 Cs) were observed at nearly equal

Journal Pre-proof

proportions in all LHP analyses (each at 20-25%). The remaining five HVS2 differences (83%) were seen in samples where the AQME read count analysis of the Ion-generated BAM files identified MM sequences with fewer consecutive Cs than the mapping-free STRait Razor analysis. Since no differences between LHP analyses were found in the Illumina datasets, which were all analyzed from the FASTQ files in CLC, the variation between the LHP analyses of the SA Ion data was likely a result of the Ion Suite Software's handling of the FASTQ data and generation of the BAM file that was used in the AQME read count analysis. It is possible that the Ion Suite Software applied preferential mapping of reads with fewer consecutive Cs to generate the BAM files analyzed in this study.

To investigate this further, the SA Ion FASTQ files were mapped to the rCRS in CLC using similar parameters as those used for the Illumina data (Supplemental File 3). In the CLC mapping of the SA Ion FASTQ files, the only difference observed between the AQME analysis of the CLC mapping and the sequence-based STRait Razor MM was the HVS2 C-tract of Sample 3 (12 and 11 consecutive Cs, respectively). STRait Razor analysis of extracted reads from the sample's Ion BAM identified a MM of 11 Cs, whereas the CLC mapped reads produced a MM with 12 Cs using STRait Razor that was consistent with the AQME analyses and other sequencing methods. Based on these results, the analysis workflow (e.g. trimming, mapping parameters) and LHP analysis tool were shown to affect the interpretation of LHP, in addition to the sequencing method. Analysis factors may explain the difference observed, including workflows (e.g. read trimming, mapping stringency, reference genome) and LHP analysis strategies (e.g. unmapped or mapped reads, read identification method, size of the region targeted, the specificity of flanking sequences). For this particular sample, the analysis workflow played a role in altering the display of LHP, and this was evident due to molecules of varying length present in relatively equal proportions. However, since the STRait Razor MM (310C 314-315-) was the outlier (all other data produced a MM haplotype of 310C 315-), it is uncertain whether the analysis accurately represented the length molecules present in the sample. Based on the comparison of different mapping workflows and LHP analysis tools, the Ion BAM files were determined to be biased towards shorter C-tracts, particularly in the HVS2 region. The bioinformatic tool, input data (e.g. FASTQ or mapped reads), and interpretation of the output may alter the LHP, which may ultimately impact the MM determination.

## 3. Conclusions

The potential for variation in the display of LHP is important to consider as mtDNA interpretation guidelines may need to be revisited with the implementation of NGS. Data generated across laboratories will undoubtedly utilize different enrichment methods, sequencing platforms, and analysis software/workflows. In this preliminary study, each of these variables was shown to have an impact on the presentation and interpretation of LHP. Although the sensitivity of NGS may generally enhance detection of heteroplasmy, damage, and mixtures [62,68,69], it may add complexity to the interpretation of LHP. LHP in the C-tracts was shown here to be inconsistent across sequencing platforms and analyses. The data from this study indicate that any C-tract with ≥7 consecutive Cs may vary in length because of the enrichment

Journal Pre-proof

method, sequencing platform, or analysis method used. CR C-tracts with ≥7 consecutive Cs are common in human mtDNA in both HVS1 (19% of global mtDNA haplotypes exhibit T16189C and thus a stretch of 10 Cs) and HVS2 (nearly all haplotypes have seven Cs present at nps 303-309). Therefore, the direct comparison of LHP between technologies is hampered.

Although the sample size was limited, and the repeatability of the data remains to be shown, the present study offers tentative updated guidelines for LHP interpretation with the introduction of NGS to forensic mtDNA analysis. Overall, caution should be exercised when evaluating C-tract indels in HVS regions, with exception to 315.1C that represents a stable insertion relative to the rCRS [55]. Substitutions that combine the HVS1 and HVS2 C-tracts (e.g. T16189C, T310C) may be called, as these were shown to be stable across sequencing platforms and analytical methods. Laboratories and databases may choose to keep C-tract indels in profiles to denote the presence of LHP; however, it is important that C-tract indels are ignored for queries and comparisons. Very few databases ignore common insertions (e.g. 16193.1C, 309.1C) and deletions at the same positions (e.g. 16193-, 309-) or other indels in the same regions (e.g. 16191.1C, 315-) [13]. If this "ignore" feature is not enabled in a database search, C-tract length variants may result in false exclusions. For instance, SA PCR enrichment (e.g. the Precision ID mtDNA Panels) and Ion sequencing of a degraded sample from a missing individual may have C-tract MM differences when compared to a high-quality reference processed with LR PCR enrichment and Illumina sequencing. The generalizability of the results from the C-tracts of HVS1 and HVS2 to the remainder of the mitogenome homopolymer regions is uncertain. Further investigations are required to determine whether the same degree of variability is observed in other LHP regions (Table S1), especially non-homopolymeric stretches such as the AC (nps 515-524) and 9 bp (nps 8281-8289) repeat regions.

The application of NGS is appealing to the forensic community due to its sensitivity and throughput, but NGS itself is not a single system. Its implementation introduces many options for enrichment, sequencing, and data analysis. For most mtDNA sequencing applications, the versatility that NGS offers is a benefit. However, for the standardization of mtDNA profile reporting of C-tracts in the commonly interrogated CR, this diverse set of options with NGS leads to diverse LHP data from the same samples. The results of this study call for further investigation into the stability of LHP, both within and across methodological workflows, as well as the interpretation of LHP in forensic genetics.

Journal Pre-proof

Journal Pre-proof

**References**

[1] K.E. Bendall, B.C. Sykes, Length heteroplasmy in the first hypervariable segment of the human mtDNA control region, Am.J.Hum.Genet. 57 (1995) 248-256.

[2] J.E. Stewart, C.L. Fisher, P.J. Aagaard, M.R. Wilson, A.R. Isenberg, D. Polanskey, et al., Length variation in HV2 of the human mitochondrial DNA control region, J.Forensic Sci. 46 (2001) 862-870.

[3] T. Melton, Mitochondrial DNA Heteroplasmy, Forensic Sci.Rev. 16 (2004) 1-20.

[4] J.A. Irwin, J.L. Saunier, H. Niederstatter, K.M. Strouss, K.A. Sturk, T.M. Diegoli, et al., Investigation of heteroplasmy in the human mitochondrial DNA control region: a synthesis of observations from more than 5000 global population samples, J.Mol.Evol. 68 (2009) 516-527.

[5] W. Parson, A. Dur, EMPOP--a forensic mtDNA database, Forensic.Sci.Int.Genet. 1 (2007) 88-92.

[6] A. Torroni, T.G. Schurr, M.F. Cabell, M.D. Brown, J.V. Neel, M. Larsen, et al., Asian affinities and continental radiation of the four founding Native American mtDNAs, Am.J.Hum.Genet. 53 (1993) 563-590.

[7] M. Tanaka, V.M. Cabrera, A.M. Gonzalez, J.M. Larruga, T. Takeyasu, N. Fuku, et al., Mitochondrial genome variation in eastern Asia and the peopling of Japan, Genome Res. 14 (2004) 1832-1850.

[8] M. Derenko, B. Malyarchuk, G. Denisova, M. Perkova, U. Rogalla, T. Grzybowski, et al., Complete mitochondrial DNA analysis of eastern Eurasian haplogroups rarely found in populations of northern Asia and eastern Europe, PLoS One. 7 (2012) e32179.

[9] A. Achilli, U.A. Perego, H. Lancioni, A. Olivieri, F. Gandini, B. Hooshiar Kashani, et al., Reconciling migration models to the Americas with the variation of North American native mitogenomes, Proc.Natl.Acad.Sci.U.S.A. 110 (2013) 14308-14313.

[10] A. Achilli, U.A. Perego, C.M. Bravi, M.D. Coble, Q.P. Kong, S.R. Woodward, et al., The phylogeny of the four pan-American MtDNA haplogroups: implications for evolutionary and disease studies, PLoS One. 3 (2008) e1764.

[11] R.S. Just, M.K. Scheible, S.A. Fast, K. Sturk-Andreaggi, A.W. Rock, J.M. Bush, et al., Full mtGenome reference data: development and characterization of 588 forensic-quality haplotypes representing three U.S. populations, Forensic.Sci.Int.Genet. 14 (2015) 141-155.

[12] A. Ramos, C. Santos, L. Mateiu, M. Gonzalez Mdel, L. Alvarez, L. Azevedo, et al., Frequency and pattern of heteroplasmy in the complete human mitochondrial genome, PLoS One. 8 (2013) e74636.

Journal Pre-proof

[13] N. Huber, W. Parson, A. Dur, Next generation database search algorithm for forensic mitogenome analyses, Forensic.Sci.Int.Genet. 37 (2018) 204-214.

[14] R.A. Ganai, E. Johansson, DNA Replication-A Matter of Fidelity, Mol.Cell. 62 (2016) 745-755.

[15] W.W. Hauswirth, M.J. Van de Walle, P.J. Laipis, P.D. Olivo, Heterogeneous mitochondrial DNA D-loop sequences in bovine tissue, Cell. 37 (1984) 1001-1007.

[16] K.A. Eckert, T.A. Kunkel, DNA polymerase fidelity and the polymerase chain reaction, PCR Methods Appl. 1 (1991) 17-24.

[17] S. Lutz, H.J. Weisser, J. Heizmann, S. Pollak, Location and frequency of polymorphic positions in the mtDNA control region of individuals from Germany, Int.J.Legal Med. 111 (1998) 67-77.

[18] W. Parson, T.J. Parsons, R. Scheithauer, M.M. Holland, Population data for 101 Austrian Caucasian mitochondrial DNA d-loop sequences: application of mtDNA sequence analysis to a forensic case, Int.J.Legal Med. 111 (1998) 124-132.

[19] P.S. Walsh, N.J. Fildes, R. Reynolds, Sequence analysis and characterization of stutter products at the tetranucleotide repeat locus vWA, Nucleic Acids Res. 24 (1996) 2807-2812.

[20] R. Szibor, I. Plate, M. Heinrich, M. Michael, R. Schoning, H. Wittig, et al., Mitochondrial D-loop (CA)n repeat length heteroplasmy: frequency in a German population sample and inheritance studies in two pedigrees, Int.J.Legal Med. 121 (2007) 207-213.

[21] G.G. Paneto, L.V. Longo, J.A. Martins, M.A. de Camargo, J.C. Costa, A.C. de Mello, et al., Heteroplasmy in hair: study of mitochondrial DNA third hypervariable region in hair and blood samples, J.Forensic Sci. 55 (2010) 715-718.

[22] P. McInerney, P. Adams, M.Z. Hadi, Error Rate Comparison during Polymerase Chain Reaction by DNA Polymerase, Mol.Biol.Int. 2014 (2014) 287430.

[23] V. Potapov, J.L. Ong, Examining Sources of Error in PCR by Single-Molecule Sequencing, PLoS One. 12 (2017) e0169774.

[24] S.B. Seo, B.S. Jang, A. Zhang, J.A. Yi, H.Y. Kim, S.H. Yoo, et al., Alterations of length heteroplasmy in mitochondrial DNA under various amplification conditions, J.Forensic Sci. 55 (2010) 719-722.

[25] M. Li, A. Schonberg, M. Schaefer, R. Schroeder, I. Nasidze, M. Stoneking, Detecting heteroplasmy from high-throughput sequencing of complete human mitochondrial DNA genomes, Am.J.Hum.Genet. 87 (2010) 237-249.

Journal Pre-proof

[26] C.Y. Pai, L.L. Hsieh, T.C. Lee, S.B. Yang, J. Linville, S.L. Chou, et al., Mitochondrial DNA sequence alterations observed between blood and buccal cells within the same individuals having betel quid (BQ)-chewing habit, Forensic Sci.Int. 156 (2006) 124-130.

[27] C.D. Calloway, R.L. Reynolds, G.L. Herrin Jr, W.W. Anderson, The frequency of heteroplasmy in the HVII region of mtDNA differs across tissue types and increases with age, Am.J.Hum.Genet. 66 (2000) 1384-1397.

[28] H. Pfeiffer, S. Lutz-Bonengel, S. Pollak, R. Fimmers, M.P. Baur, B. Brinkmann, Mitochondrial DNA control region diversity in hairs and body fluids of monozygotic triplets, Int.J.Legal Med. 118 (2004) 71-74.

[29] S. Lutz, H.J. Weisser, J. Heizmann, S. Pollak, Mitochondrial heteroplasmy among maternally related individuals, Int.J.Legal Med. 113 (2000) 155-161.

[30] C. Bini, G. Pappalardo, mtDNA HVI length heteroplasmic profile in different tissues of maternally related members, Forensic Sci.Int. 152 (2005) 35-38.

[31] M. Asari, J. Azumi, K. Shimizu, H. Shiono, Differences in tissue distribution of HV2 length heteroplasmy in mitochondrial DNA between mothers and children, Forensic Sci.Int. 175 (2008) 155-159.

[32] L. Forster, P. Forster, S.M. Gurney, M. Spencer, C. Huang, A. Rohl, et al., Evaluating length heteroplasmy in the human mitochondrial DNA control region, Int.J.Legal Med. 124 (2010) 133-142.

[33] W. Parson, L. Gusmao, D.R. Hares, J.A. Irwin, W.R. Mayr, N. Morling, et al., DNA Commission of the International Society for Forensic Genetics: revised and extended guidelines for mitochondrial DNA typing, Forensic.Sci.Int.Genet. 13 (2014) 134-142.

[34] N.J. Loman, R.V. Misra, T.J. Dallman, C. Constantinidou, S.E. Gharbia, J. Wain, et al., Performance comparison of benchtop high-throughput sequencing platforms, Nat.Biotechnol. (2012).

[35] M. Mikkelsen, R. Frank-Hansen, A.J. Hansen, N. Morling, Massively parallel pyrosequencing of the mitochondrial genome with the 454 methodology in forensic genetics, Forensic.Sci.Int.Genet. 12 (2014) 30-37.

[36] M.A. Quail, M. Smith, P. Coupland, T.D. Otto, S.R. Harris, T.R. Connor, et al., A tale of three next generation sequencing platforms: comparison of Ion Torrent, Pacific Biosciences and Illumina MiSeq sequencers, BMC Genomics. 13 (2012) 341.

[37] M.G. Ross, C. Russ, M. Costello, A. Hollinger, N.J. Lennon, R. Hegarty, et al., Characterizing and measuring bias in sequence data, Genome Biol. 14 (2013) R51.

Journal Pre-proof

[38] C. Davis, D. Peters, D. Warshauer, J. King, B. Budowle, Sequencing the hypervariable regions of human mitochondrial DNA using massively parallel sequencing: Enhanced data acquisition for DNA samples encountered in forensic testing, Leg.Med.(Tokyo). 17 (2015) 123-127.

[39] C.Y. Lin, L.C. Tsai, H.M. Hsieh, C.H. Huang, Y.J. Yu, B. Tseng, et al., Investigation of length heteroplasmy in mitochondrial DNA control region by massively parallel sequencing, Forensic.Sci.Int.Genet. 30 (2017) 127-133.

[40] M.A. Peck, K. Sturk-Andreaggi, J.T. Thomas, R.S. Oliver, S.M. Barritt-Ross, C.K. Marshall, Developmental Validation of a Nextera XT Mitogenome Illumina MiSeq Sequencing Method for High Quality Samples, Forensic Sci. Int. Gene. 34 (2018) 25-36.

[41] W. Parson, C. Strobl, G. Huber, B. Zimmermann, S.M. Gomes, L. Souto, et al., Evaluation of next generation mtGenome sequencing using the Ion Torrent Personal Genome Machine (PGM), Forensic.Sci.Int.Genet. 7 (2013) 543-549.

[42] C. Strobl, M. Eduardoff, M.M. Bus, M. Allen, W. Parson, Evaluation of the precision ID whole MtDNA genome panel for forensic analyses, Forensic.Sci.Int.Genet. 35 (2018) 21-25.

[43] V. Pereira, A. Longobardi, C. Borsting, Sequencing of mitochondrial genomes using the Precision ID mtDNA Whole Genome Panel, Electrophoresis. 39 (2018) 2766-2775.

[44] A.E. Woerner, A. Ambers, F.R. Wendt, J.L. King, R.S. Moura-Neto, R. Silva, et al., Evaluation of the precision ID mtDNA whole genome panel on two massively parallel sequencing systems, Forensic.Sci.Int.Genet. 36 (2018) 213-224.

[45] L. Chaitanya, A. Ralf, M. van Oven, T. Kupiec, J. Chang, R. Lagace, et al., Simultaneous Whole Mitochondrial Genome Sequencing with Short Overlapping Amplicons Suitable for Degraded DNA Using the Ion Torrent Personal Genome Machine, Hum.Mutat. 36 (2015) 1236-1247.

[46] W. Parson, G. Huber, L. Moreno, M.B. Madel, M.D. Brandhagen, S. Nagl, et al., Massively parallel sequencing of complete mitochondrial genomes from hair shaft samples, Forensic.Sci.Int.Genet. 15 (2015) 8-15.

[47] J.L. King, B.L. LaRue, N.M. Novroski, M. Stoljarova, S.B. Seo, X. Zeng, et al., High-quality and high-throughput massively parallel sequencing of the human mitochondrial genome using the Illumina MiSeq, Forensic.Sci.Int.Genet. 12 (2014) 128-135.

[48] M.A. Peck, M.D. Brandhagen, C. Marshall, T.M. Diegoli, J.A. Irwin, K. Sturk-Andreaggi, Concordance and reproducibility of a next generation mtGenome sequencing method for high-quality samples using the Illumina MiSeq, Forensic.Sci.Int.Genet. 24 (2016) 103-111.

Journal Pre-proof

[49] S. Riman, K.M. Kiesler, L.A. Borsuk, P.M. Vallone, Characterization of NIST human mitochondrial DNA SRM-2392 and SRM-2392-I standard reference materials by next generation sequencing, Forensic.Sci.Int.Genet. 29 (2017) 181-192.

[50] J.A. McElhoe, M.M. Holland, K.D. Makova, M.S. Su, I.M. Paul, C.H. Baker, et al., Development and assessment of an optimized next-generation DNA sequencing approach for the mtgenome using the Illumina MiSeq, Forensic.Sci.Int.Genet. 13 (2014) 20-29.

[51] M. Scheible, R. Just, K. Sturk-Andreaggi, J. Saunier, W. Parson, T. Parsons, et al., The mitochondrial landscape of African Americans: An examination of more than 2500 control region haplotypes from 22 U.S. locations, Forensic.Sci.Int.Genet. 22 (2016) 139-148.

[52] A. Brandstatter, C.T. Peterson, J.A. Irwin, S. Mpoke, D.K. Koech, W. Parson, et al., Mitochondrial DNA control region sequences from Nairobi (Kenya): inferring phylogenetic parameters for the establishment of a forensic database, Int.J.Legal Med. 118 (2004) 294-306.

[53] J.A. Irwin, J.L. Saunier, K.M. Strouss, K.A. Sturk, T.M. Diegoli, R.S. Just, et al., Development and expansion of high-quality control region databases to improve forensic mtDNA evidence interpretation, Forensic.Sci.Int.Genet. 1 (2007) 154-157.

[54] S. Anderson, A.T. Bankier, B.G. Barrell, M.H. de Bruijn, A.R. Coulson, J. Drouin, et al., Sequence and organization of the human mitochondrial genome, Nature. 290 (1981) 457-465.

[55] R.M. Andrews, I. Kubacka, P.F. Chinnery, R.N. Lightowlers, D.M. Turnbull, N. Howell, Reanalysis and revision of the Cambridge reference sequence for human mitochondrial DNA, Nat.Genet. 23 (1999) 147.

[56] J.D. Ring, K. Sturk-Andreaggi, M.A. Peck, C.K. Marshall, A Performance Evaluation of Nextera XT and KAPA HyperPlus for Rapid Illumina Library Preparation of Long-Range Mitogenome Amplicons, Forensic Sci.Int.Genet. 29 (2017) 174-180.

[57] K. Sturk-Andreaggi, M.A. Peck, C. Boysen, P. Dekker, T.P. McMahon, C.K. Marshall, AQME: A forensic mitochondrial DNA analysis tool for next-generation sequencing data, Forensic Sci.Int.Genet. 31 (2017) 189-197.

[58] M. van Oven, PhyloTree Build 17: Growing the human mitochondrial DNA tree, Forensic Science International: Genetics Supplement Series. 5 (2015) e392-e394.

[59] M. van Oven, M. Kayser, Updated comprehensive phylogenetic tree of global human mitochondrial DNA variation, Hum.Mutat. 30 (2009) E386-94.

[60] Scientific Working Group on DNA Analysis Methods (SWGDAM), Guidelines for mitochondrial DNA (mtDNA) nucleotide sequence interpretation, Forensic Sci Comm. 5 (2003) 1-5.

Journal Pre-proof

[61] J.D. Ring, K. Sturk-Andreaggi, M.A. Peck, C. Marshall, Bioinformatic removal of NUMT-associated variants in mitotiling next-generation sequencing data from whole blood samples, Electrophoresis. 39 (2018) 2785-2797.

[62] R.S. Just, J.A. Irwin, W. Parson, Mitochondrial DNA heteroplasmy in the emerging field of massively parallel sequencing, Forensic.Sci.Int.Genet. 18 (2015) 131-139.

[63] D.H. Warshauer, D. Lin, K. Hari, R. Jain, C. Davis, B. Larue, et al., STRait Razor: a length-based forensic STR allele-calling tool for use with second generation sequencing data, Forensic.Sci.Int.Genet. 7 (2013) 409-417.

[64] A.E. Woerner, J.L. King, B. Budowle, Fast STR allele identification with STRait Razor 3.0, Forensic.Sci.Int.Genet. 30 (2017) 18-23.

[65] J.L. King, F.R. Wendt, J. Sun, B. Budowle, STRait Razor v2s: Advancing sequence-based STR allele reporting and beyond to other marker systems, Forensic.Sci.Int.Genet. 29 (2017) 21-28.

[66] D.H. Warshauer, J.L. King, B. Budowle, STRait Razor v2.0: the improved STR Allele Identification Tool--Razor, Forensic.Sci.Int.Genet. 14 (2015) 182-186.

[67] S. Lutz-Bonengel, T. Sanger, S. Pollak, R. Szibor, Different methods to determine length heteroplasmy within the mitochondrial control region, Int.J.Legal Med. 118 (2004) 274-281.

[68] M.M. Holland, M.R. McQuillan, K.A. O'Hanlon, Second generation sequencing allows for mtDNA mixture deconvolution and high resolution detection of heteroplasmy, Croat.Med.J. 52 (2011) 299-313.

[69] M.M. Holland, E.D. Pack, J.A. McElhoe, Evaluation of GeneMarker(R) HTS for improved alignment of mtDNA MPS data, haplotype determination, and heteroplasmy assessment, Forensic.Sci.Int.Genet. 28 (2017) 90-98.

Journal Pre-proof